UDS
unmanned defense systems

AVENGER 5

Preparation and Safety Instructions

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Systematic Approach to High-Tech Powered Defense

PROPRIETARY INFORMATION

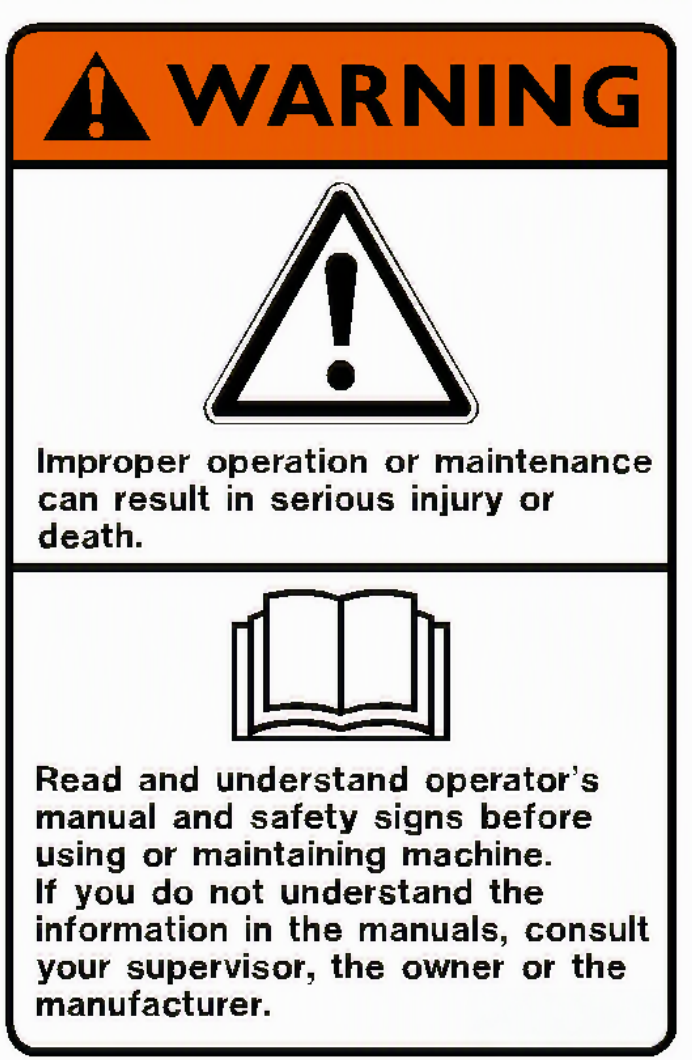
WARNING - Read Before Operation!

Read and understand these Preparation and Safety Instructions and all related user manuals in full before assembling, preparing, handling, powering on, or operating the UAV system.

Do not proceed unless you have received the required training and fully understand all procedures, warnings, and safety requirements.

Additional user manuals:

1. AVENGER 5 User Manual
2. UAV launcher PL-40
3. Warheaddocumentation
4. ESAD documentation
5. Applicable storage, transportation, handling, and maintenance documentation
6. Operating-base procedures and all other applicable safety and regulatory documentation



PROPRIETARY INFORMATION

WARNING - Warhead Storage and Transportation

Before installing the warhead in the UAV, ensure that all applicable storage, transportation, handling, and transfer procedures have been completed in accordance with the regulations and safety requirements of the operating base.

Warhead handling and installation may be performed only by appropriately trained and authorised personnel.



PROPRIETARY INFORMATION

Set of Equipment

This document provides instructions for preparing the UAV for flight, including vehicle assembly, pre-flight checks, safety procedures, and diagnostic guidance. It is intended to support the safe and correct preparation of the system before each flight.

The UAV system set consists of two main containers:
1. UAV transport box
2. Tool box
3. Battery box

Additional Equipment:
1. Catapult launch system
2. Ground control station







The actual equipment set may differ from that shown in the illustrative images.

PROPRIETARY INFORMATION

UAV Launcher PL-40

1. Assemble and position the catapult in accordance with the Catapult User Manual.
2. Set the specified launch pressure.
3. Verify that all catapult sections, fasteners, connections, and locking mechanisms are correctly installed and firmly tightened.

WARNING - Launch-Speed Risk!

All catapult connections and locking mechanisms must be tightened firmly. Loose or incorrectly assembled components may prevent the UAV from reaching the required launch speed, potentially resulting in an unsuccessful launch or loss of the UAV.

Refer to the UAV launcher PL-40 User Manual for the complete assembly, pressure-setting, and safety procedure.



The actual equipment set may differ from that shown in the illustrative images.

PROPRIETARY INFORMATION

Ground Control Station Preparation

1. Set up the ground control station in the designated operating area.
2. Install and secure the antenna.
3. Connect and power on the tablets.
4. Calibrate the antenna in accordance with the SwarmC2 User Manual.
5. Confirm that all ground control station components are connected and operating correctly.

Once the ground control station setup is complete, continue with the UAV preparation procedure.



The actual equipment set may differ from that shown in the illustrative images.

PROPRIETARY INFORMATION

ESD grounding

After opening the UAV transport box, the operator must touch the designated ESD grounding point everytime before handling the vehicle or its components. This equalises the electrical potential between the operator and the UAV, reducing the risk of electrostatic discharge and damage to sensitive electronics.

The ESD grounding point is identified by a yellow label on the UAV.





PROPRIETARY INFORMATION

Open the Payload Bay

While the UAV remains secured inside the transport box, unlock and open the payload bay.





PROPRIETARY INFORMATION

Warhead Installation – Part 1

1. Touch the designated ESD grounding point everytime before handling the vehicle or its components.
2. Place the Warhead in the payload bay with its top side facing upward to allow convenient access to the connections.
3. Securely connect the payload cables to the ESAD using the 1.5 mm HEX screw terminals.

WARNING - Trained Personnel Only!

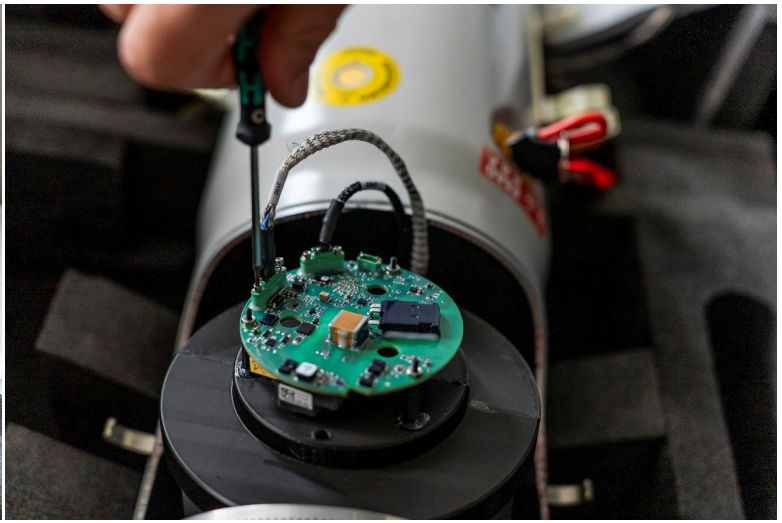
Warhead installation and handling must be performed only by personnel who have received the required training.

Do not install, handle, or operate the Warhead unless you are properly trained and authorised.









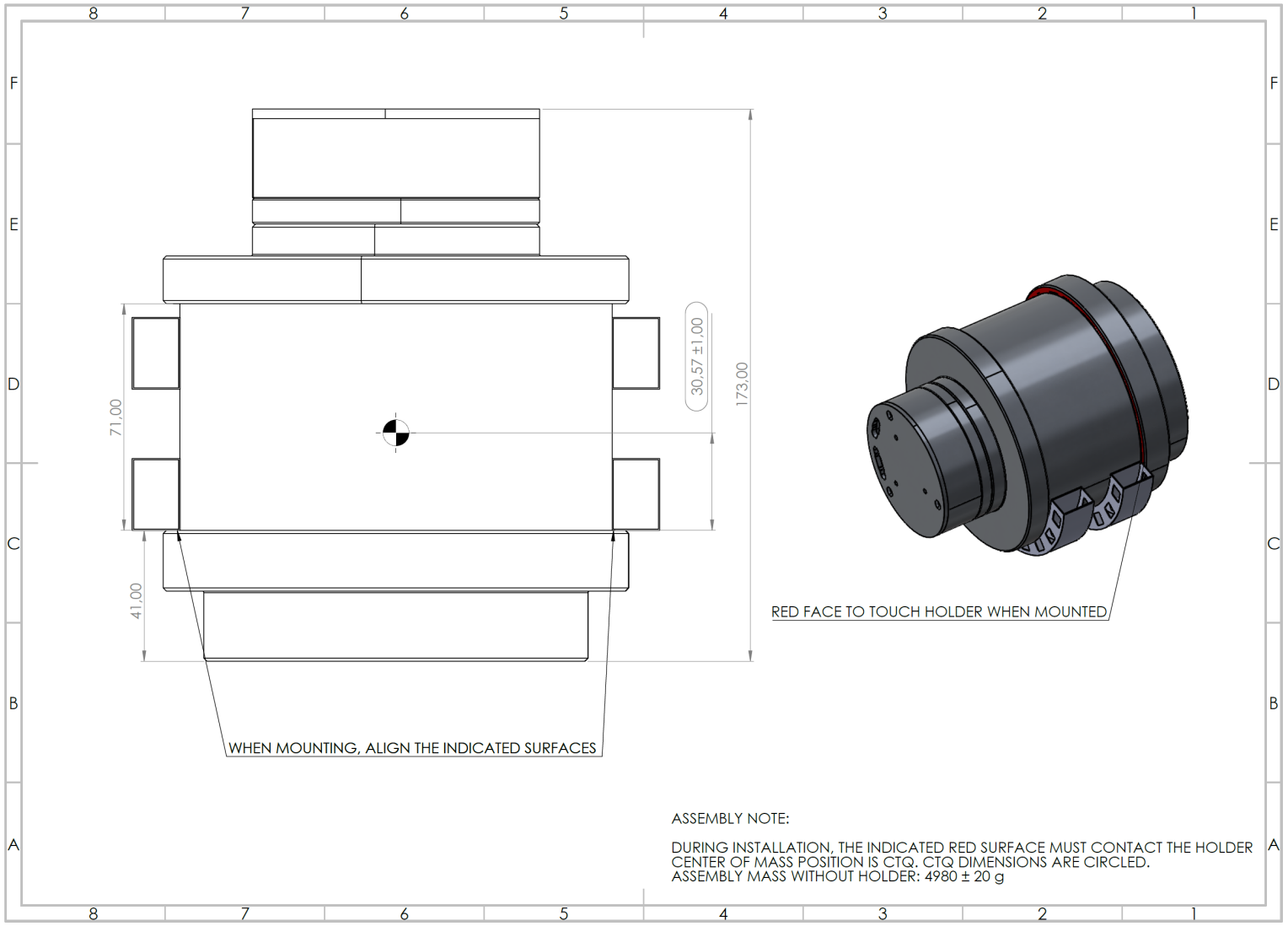
PROPRIETARY INFORMATION

Warhead Installation – Part 2

WARNING — Payload Position!

Pay close attention to the payload position and instructions in the drawing!

Due to its significant mass, incorrect placement may shift the UAV's centre of gravity and cause the aircraft to become unbalanced. Ensure that the payload is correctly seated and secured in the designated position.



PROPRIETARY INFORMATION

Warhead Installation – Part 3

1. Carefully lower the Warhead into its mounting brackets.
2. Verify that the Warhead is positioned correctly to maintain the specified UAV centre of gravity.
3. Tighten the retaining rings to secure the Warhead.
4. Close and lock the payload bay hatch.

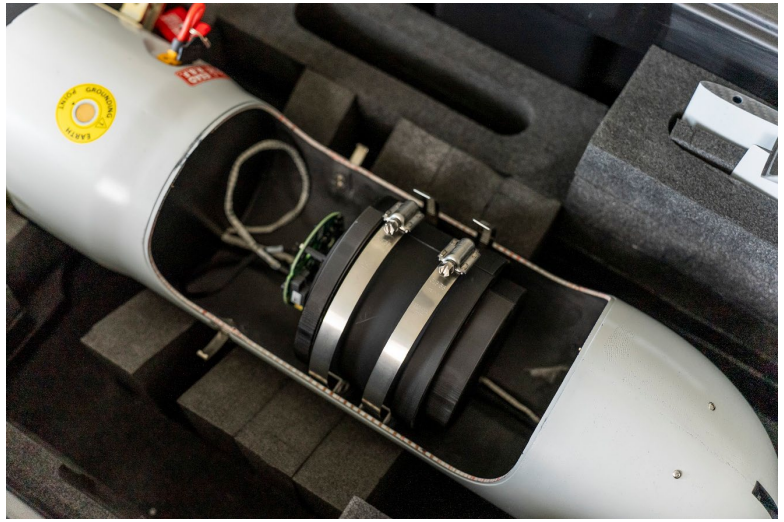
WARNING - Trained Personnel Only!

Warhead installation and handling must be performed only by personnel who have received the required training.

Do not install, handle, or operate the Warhead unless you are properly trained and authorised.

Refer to the Warhead Specification Drawing for the warhead mass, center of gravity (CG) location, and installation position.







PROPRIETARY INFORMATION

Battery Installation – Part 1

1. Open the battery transport box and visually inspect the battery for damage, swelling, deformation, leakage, or loose connectors.
2. Unlock and open the UAV battery-compartment hatch.
3. Remove the hatch and place it in a clean, safe location.
4. Before inserting the battery, connect the battery-retaining strap using its buckle. Once the battery is installed, access to the buckle will be restricted, making it difficult to connect.

WARNING - Do not use the battery if any abnormality is found!

Fully charge the battery before installing it in the airframe.

Verify that the battery identification number matches the identification number on the fuselage.

The QR codes on the battery and fuselage must also match.









PROPRIETARY INFORMATION

Battery Installation – Part 2

1. Insert the fully charged battery into the battery compartment, sliding the end without the cable in first.
2. Secure the battery firmly using the retaining strap.
3. Connect the battery connector and tighten its locking ring securely.

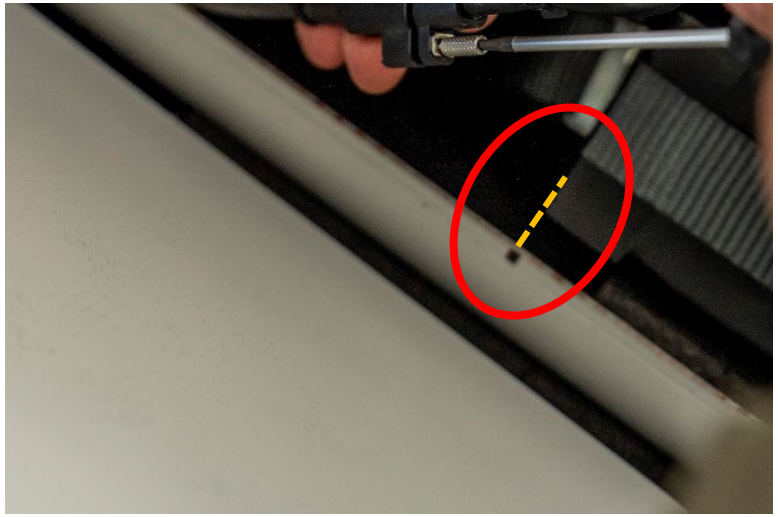
Ensure that the battery is fully inserted into the fuselage and seated firmly against the bulkhead.

Verify that the end of the battery aligns with the position marker on the fuselage.





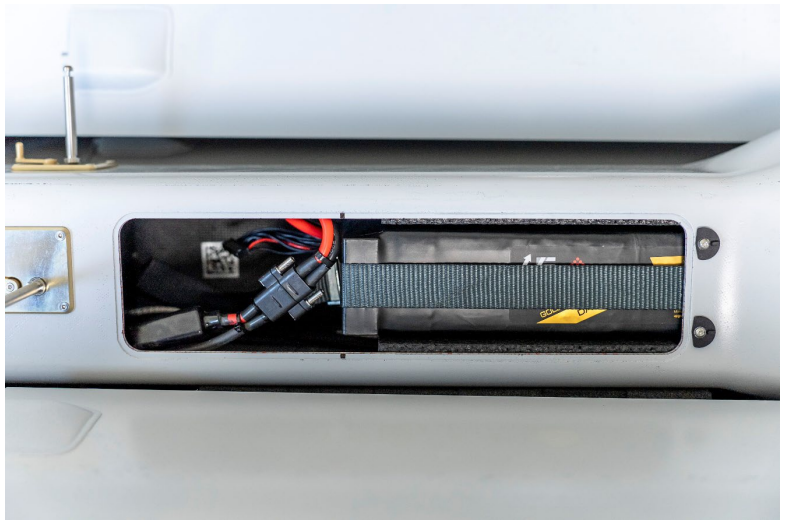




PROPRIETARY INFORMATION

Battery Installation – Part 3

1. Arrange the cable inside the battery compartment so that the connector sits flush with the surrounding surface.
2. Do not sharply bend, pinch, twist, or route the cable with a small bend radius.
3. Refit the battery-compartment hatch.
4. Secure the hatch using the locking pins.







PROPRIETARY INFORMATION

Mounting the UAV on the Catapult

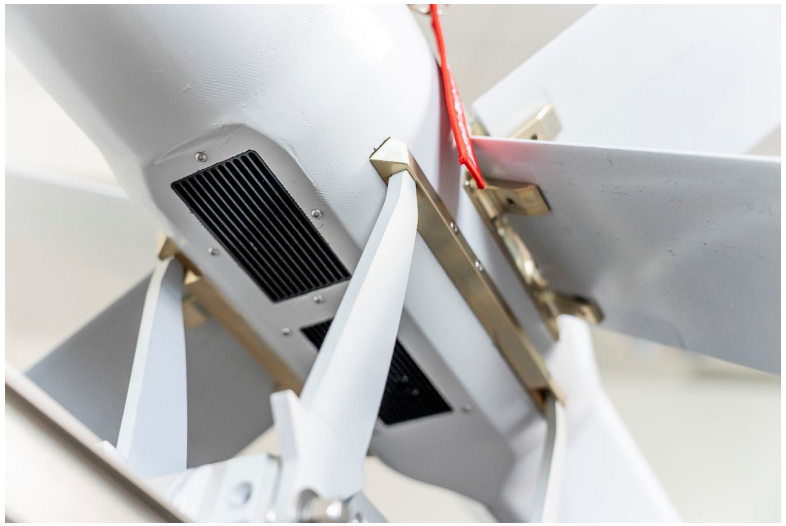
The UAV is now ready to be mounted on the catapult.

1. Keep the Remove Before Flight (RBF) pin installed.
2. Carefully lift the UAV out of the transport box (2 persons are required for this oppearation).
3. Take care not to apply pressure to or damage the wings, control surfaces, or empennage.
4. Position the UAV on the catapult trolley and ensure it is correctly seated in the designated mounting fixtures.









PROPRIETARY INFORMATION

Final Airframe Assembly

Once the UAV is correctly mounted on the catapult trolley, it is ready for final airframe assembly.



PROPRIETARY INFORMATION

Securing the Wings

1. Remove the four pre-installed M6 × 30 mm wing mounting bolts from the front mounting brackets.
2. Take the corresponding key from the utility box.
3. Carefully rotate the wing into its flight position. The wing is loosely secured by a single bolt located near the trailing edge.
4. Align and gently position the wing until it seats correctly in the mounting bracket. Do not apply excessive force.
5. Install the M6 × 30 mm mounting bolts that were removed in Step 1 into their original positions.
6. Tighten all eight wing-mounting bolts securely.
7. Verify that the wing is correctly seated, aligned, and free from movement.

WARNING - Support the UAV During Wing Installation!

While rotating the wings into position, firmly support the UAV frame against the catapult trolley. The UAV may slip out of the mounting fixtures if it is not properly supported.
Move each wing slowly and carefully. Do not apply excessive force.

It is also important not to bend, force, or push the cables into the UAV. Ensure that the cables are not pinched, and route the RF cable smoothly around the hinge.









PROPRIETARY INFORMATION

Tail Stabiliser Installation

1. Remove all four tail stabilisers from the UAV transport box. Ensure that the threaded retaining pins are pre-installed in each stabiliser but fully loosened before installation.
2. Slide each stabiliser onto its mounting shaft. The stabilisers are identical and may be installed on either side.
3. Press each stabiliser fully into position by hand. Ensure that the control horn is inserted into the stabilizer during installation.
4. Finger-tighten the retaining screws to secure the stabilisers.
5. Verify that both stabilisers are fully seated, correctly aligned, and secure.

WARNING - Support the UAV During Stabiliser Installation!

While installing the tail stabilisers, firmly support the UAV frame against the catapult trolley. The UAV may slip out of the mounting fixtures if it is not properly supported.
Install each stabiliser slowly and carefully. Do not apply excessive force.



PROPRIETARY INFORMATION

Final Mechanical Check

This is the final mechanical check before proceeding to the last preparation stage and powering up the UAV.

Ensure that both folding propeller blades are folded rearward, toward the tail of the UAV. If the blades are incorrectly positioned, they may strike the catapult trolley during launch, potentially damaging the propeller and causing loss of thrust or loss of the UAV.

WARNNING - Do not proceed with power-up until the propeller blades are correctly positioned!





PROPRIETARY INFORMATION

Removing the Front RBFs - 1

Remove the following Remove Before Flight (RBF) devices from the front of the UAV:

1. Pitot-tube cover.
2. Sensor protective cover.
3. Gimbal securing strap.

Verify that both RBF devices have been fully removed and that the pitot tube, sensor, and gimbal are unobstructed and undamaged.

When removing the RBF devices, proceed gently and support the UAV to prevent it from being pushed out of position on the catapult trolley!





PROPRIETARY INFORMATION

Removing the Power RBF - 2

1. Remove the power **Remove Before Flight (RBF)** device.
2. Close the RBF access lid.
3. Ensure that both the RBF and umbilical access lids are fully closed, correctly seated, and securely locked.

When removing the RBF devices, proceed gently and support the UAV to prevent it from being pushed out of position on the catapult trolley!

The UAV will now begin its automatic power-on sequence and establish a connection with the ground control station. Once the connection is established, follow the pre-flight checklist displayed on the ground control equipment.









PROPRIETARY INFORMATION

Initial Power-On and System Preparation

With #1 and #2 RBF devices removed and the access lids securely closed, proceed with:

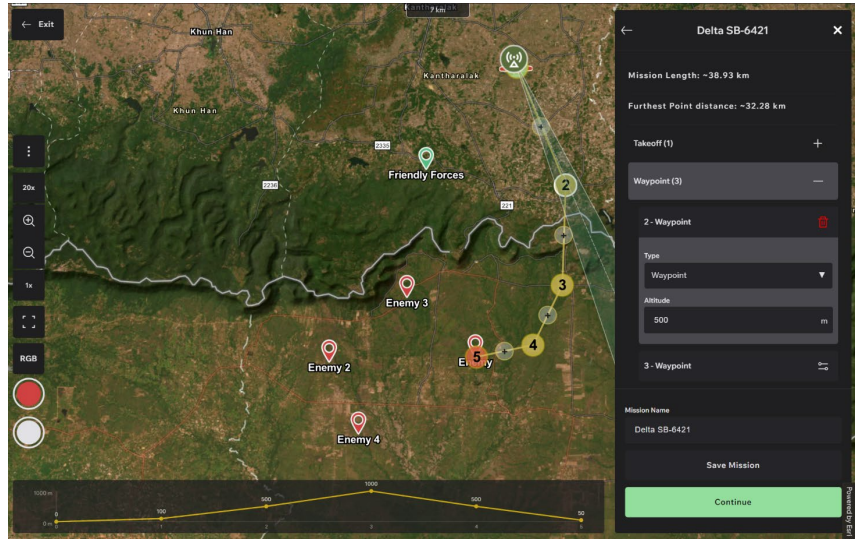
1. Initial power-on sequence.
2. Initial system diagnostics.
3. Ground-control-station checks.
4. Mission planning and mission-data verification.

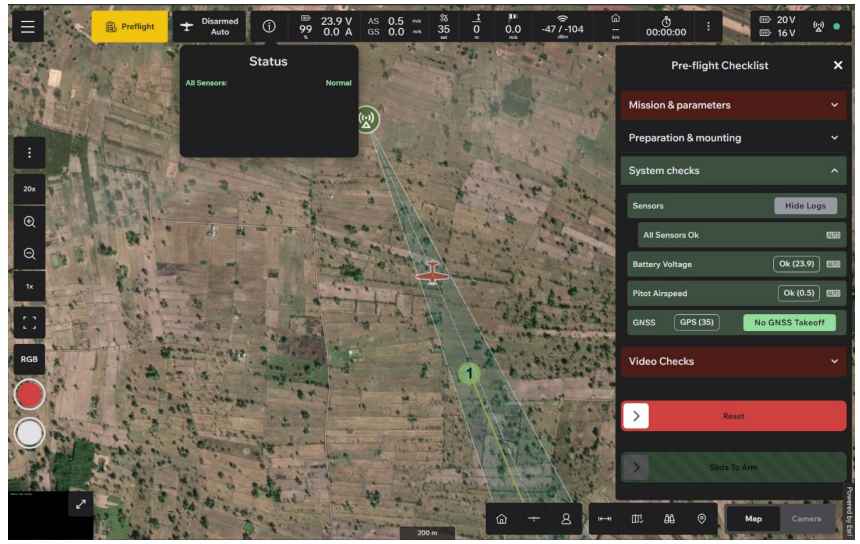
WARNING - System Status Confirmation!

Complete and confirm all checks before proceeding to the final pre-launch sequence.

Do not proceed if any system, indication, or parameter is not nominal. Resolve all identified issues before continuing.

For more detailed instructions, refer to the applicable user manuals.





WARNING — FINAL SAFETY GATE: WARHEAD RBF #3

The Warhead #3 Remove Before Flight (RBF) device is the final safety device. Remove it only when instructed by the SwarmC2 pre-flight checklist and after all of the following conditions have been met:

1. The UAV and ESAD are powered on.
2. All initial UAV and ESAD diagnostics have been completed successfully, with all indications nominal.
3. A stable connection to the UAV has been established and confirmed through SwarmC2.
4. Mission planning has been completed and verified.
5. The payload, ground control station, catapult, and launch system are fully prepared and confirmed ready.
6. All preceding preparation steps, safety checks, inspections, confirmations, and authorizations have been completed.
7. The relevant checks have been formally confirmed by appropriately qualified and authorized personnel. This includes, but is not limited to, the warhead and ESAD providers, their authorized representatives, and personnel specifically trained and certified for the relevant equipment.
8. All non-essential personnel have cleared the designated launch area.

Do not remove the ESAD RBF before the UAV and ESAD have been powered on and the initial checks have been completed. Removing it before ESAD power-up will trigger an immediate ESAD fault.

SwarmC2 will indicate when to remove the RBF, automatically verify its removal, and confirm whether the ESAD status is Ready to Fly.

When removing the RBF, proceed gently and support the UAV to prevent it from being pushed out of position on the catapult trolley.

Do not remove the Warhead RBF or proceed with launch if any required check, confirmation, or authorisation is incomplete; if any condition is not nominal; or if SwarmC2 does not confirm the ESAD as Ready to Fly.





PROPRIETARY INFORMATION

Ready for Launch

1. Confirm that the UAV, payload, ground control station, and catapult are ready for launch.
2. Ensure that all personnel have moved to the designated safe distance and that the launch area is clear.
3. From the designated safe position, perform the final launch confirmation.
4. Launch the UAV in accordance with the approved launch procedure.

WARNING - Launch Area!

Do not initiate the launch until all personnel are clear of the catapult and the launch path, and the launch area has been confirmed safe.



PROPRIETARY INFORMATION

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